

W3

Threat Detection Platform

4
DAYS

Project Information

Project	Threat Detection Platform
Week	Week 3 — Alerting + Response Automation
Data Flow	Python + MongoDB + Threat Intelligence
Stack	Python + MongoDB + Flask

Week 3 Objectives

By the end of Week 3, the Threat Detection Platform should be able to:

- generate alerts for suspicious threats
- identify HIGH and MEDIUM risk IPs
- log security alerts into MongoDB
- simulate automated response actions
- create a basic SOC-style monitoring workflow
- prepare alert data for dashboard monitoring

DAY 11

Alert Generation Engine

STEP 1 — Create Alert Engine Module

(Goal: Generate alerts for suspicious threats)

```
from db import collection
def generate_alerts():
    threats = collection.find({
        "risk_level": {"$in": ["HIGH", "MEDIUM"]}
    })

    alerts = []

    for threat in threats:
        alert = {
            "ip": threat["ip"],
```

```
        "risk_level": threat["risk_level"],
        "message": f"Suspicious activity detected from {threat['ip']}"
        ({threat['risk_level']})"
    }

    alerts.append(alert)
    print(alert)

    return alerts

generate_alerts()
```

Expected Output

```
{'ip': '8.8.8.8', 'risk_level': 'HIGH', 'message': 'Suspicious activity
detected from 8.8.8.8 (HIGH)'}
```

DAY 12

Alert Logging System

STEP 2 — Create Alert Logging Module (Goal: Store alerts in MongoDB)

```
from db import db
from alert_engine import generate_alerts
from datetime import datetime

alert_collection = db["alerts"]

def store_alerts():
    alerts = generate_alerts()

    for alert in alerts:
        alert["timestamp"] = datetime.utcnow()

        alert_collection.insert_one(alert)

        print(f"Stored alert for {alert['ip']}")

store_alerts()
```

Expected Output:

```
Stored alert for 8.8.8.8
```

STEP 3 — Create Response Automation Module
(Goal: Simulate automated SOC response)

```
from db import collection

def automated_response():
    threats = collection.find({
        "risk_level": "HIGH"
    })

    for threat in threats:
        print(f"[ACTION] Blocking suspicious IP: {threat['ip']}
({threat['risk_level']})")

automated_response()
```

Expected Output:

```
[ACTION] Blocking suspicious IP: 8.8.8.8 (HIGH)
```

STEP 4 — Execute Week 3 Modules
(Goal: Validate alerting workflow)

```
python alert_engine.py
python log_alerts.py
python response_action.py
```

Expected Output:

```
Stored alert for 8.8.8.8
[ACTION] Blocking suspicious IP: 8.8.8.8 (HIGH)
```

Completion Checklist

- ✓ Alert engine created
- ✓ Suspicious IP detection implemented
- ✓ Alert logging system completed
- ✓ MongoDB alert storage working
- ✓ Automated response simulation added
- ✓ Week 3 pipeline executed successfully
- ✓ SOC-style monitoring workflow created

Week 3 — Required GitHub Commits

```
git commit -m "feat: add alert generation engine"  
git commit -m "feat: add alert logging system"  
git commit -m "feat: add automated response actions"  
git commit -m "test: verify week 3 alert pipeline"
```